

BG77xA-GL&BG95xA-GL& BG950S-GL

QCFG AT Commands Manual

LPWA Module Series

Version: 1.1

Date: 2024-09-11

Status: Released



At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

<http://www.quectel.com/support/sales.htm>.

For technical support, or to report documentation errors, please visit:

<http://www.quectel.com/support/technical.htm>.

Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an "as available" basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

- a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
- b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
- c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
- d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2024. All rights reserved.

About the Document

Revision History

Version	Date	Author	Description
-	2021-01-29	Igor STANCIC/ Aleksandar PAUNOVIC/ Mladen VUKOVIC	Creation of the document
1.0	2022-05-31	Igor STANCIC/ Aleksandar PAUNOVIC/ Mladen VUKOVIC	First official release
1.1	2024-09-11	Nikola PERIC/ Milos PJANOVIC	- Numerous changes were made to this document. It should be read in its entirety. - Added the applicable module BG950S-GL. - Deleted AT+QCFG="setcfg". - Updated AT+QCFG (Chapter 2.1). - Added AT+QCFG="bandrestore" (Chapter 2.1.1.2). - Added AT+QCFG="epco" (Chapter 2.1.1.4). - Added AT+QCFG="ntpsync" (Chapter 2.1.1.9). - Added AT+QCFG="extmobility" (Chapter 2.1.1.10). - Added AT+QCFG="sibinfo" (Chapter 2.1.1.14). - Added AT+QCFG="emmtimer" (Chapter 2.1.1.15). - Added AT+QCFG="emmcause" (Chapter 2.1.16). - Added AT+QCFG="raiflag" (Chapter 2.1.1.17). - Added AT+QCFG="autoconnect" (Chapter 2.1.1.18). - Added AT+QCFG="automatic_recovery" (Chapter 2.1.2.4). - Added AT+QCFG="autopwrkey" (Chapter 2.1.2.9). - Updated general description for AT+QCFG="nwoper" (Chapter 2.1.2.12). - Added AT+QCFG="simcfg" (Chapter 2.1.2.16). - Added AT+QCFG="simcmd" (Chapter 2.1.2.17).

-
19. Added AT+QCFG="sim_status_check" (Chapter 2.1.2.18).
 20. Updated the <value> parameter value for AT+QCFG="stored/cell" (Chapter 2.1.2.19).
 21. Updated the <enable> parameter value for AT+QCFG="usb" (Chapter 2.1.2.26).
 22. Added AT+QCFG="cmux/urcport" (Chapter 2.1.2.28).
 23. Added AT+QCFG="fast/poweroff" (Chapter 2.1.2.29).
 24. Added AT+QCFG="dhclkrefresh" (Chapter 2.1.2.30).
-

Contents

About the Document.....	3
Contents	5
Table Index.....	7
1 Introduction	8
1.1. Definitions	8
1.2. AT Command Syntax.....	8
1.3. AT Command Responses.....	9
1.4. Declaration of AT Command Examples	10
2 Description of AT+QCFG Commands.....	11
2.1. AT+QCFG Extended Configuration Settings.....	11
2.1.1. Network-Related AT Commands.....	13
2.1.1.1. AT+QCFG="band" Set Frequency Band.....	13
2.1.1.2. AT+QCFG="bandrestore" Restore Default Band Configuration	15
2.1.1.3. AT+QCFG="celevel" Query NB-IoT Coverage Enhancement Level	15
2.1.1.4. AT+QCFG="epco" Set PCO Encoding Type: PCO or ePCO.....	16
2.1.1.5. AT+QCFG="nwscanmode" Set RAT(s) to be Searched for	17
2.1.1.6. AT+QCFG="iotopmode" Set Network Category to be Searched Under LTE RAT	18
2.1.1.7. AT+QCFG="lapiconf" Set LAPM Value (Low Access Priority Indicator)	19
2.1.1.8. AT+QCFG="nwscanseq" Set RATs Searching Sequence	20
2.1.1.9. AT+QCFG="ntpsync" Synchronize Time in Milliseconds with NTP Server	21
2.1.1.10. AT+QCFG="extmobility" Set Mobility Type	21
2.1.1.11. AT+QCFG="psm/enter" Trigger Module into PSM Immediately	22
2.1.1.12. AT+QCFG="psm/urc" Enable/Disable PSM Entering Indication.....	23
2.1.1.13. AT+QCFG="servicedomain" Set Service Domain	24
2.1.1.14. AT+QCFG="sibinfo" Query SIB Information.....	25
2.1.1.15. AT+QCFG="emmtimer" Query EMM Timer	26
2.1.1.16. AT+QCFG="emmcause" Get EMM Cause Value	26
2.1.1.17. AT+QCFG="raiflag" Set RAI Flag.....	27
2.1.1.18. AT+QCFG="autoconnect" Enable/Disable Auto Connect Mode.....	29
2.1.2. Platform-Related AT Commands	30
2.1.2.1. AT+QCFG="airplanecontrol" Enable/Disable Airplane Mode Control via W_DISABLE# ..	30
2.1.2.2. AT+QCFG="apn/display" Enable/Disable Showing APN	32
2.1.2.3. AT+QCFG="apready" Set AP_READY Behavior	33
2.1.2.4. AT+QCFG="automatic_recovery" Enable/Disable Automatic Recovery	34
2.1.2.5. AT+QCFG="edrxusimact" Activate/Deactivate (U)SIM in eDRX Mode	35
2.1.2.6. AT+QCFG="emux/urcport" Set URC Output Port to EMUX Mode	36
2.1.2.7. AT+QCFG="gpio" Set GPIO Status	36
2.1.2.8. AT+QCFG="lte/bandprior" Set Band Scan Priority under LTE	38
2.1.2.9. AT+QCFG="autopwrkey" Enable/Disable Autoboot.....	39
2.1.2.10. AT+QCFG="lwm2m" Enable/Disable LwM2M	40
2.1.2.11. AT+QCFG="netupd" Enable/Disable Network Override for Specified LTE Parameters...	41

2.1.2.12. AT+QCFG="nwoper"	Set Network Operator Mode of Modem.....	42
2.1.2.13. AT+QCFG="ps_dev_mob_type"	Set Device Mobility Type Flag	43
2.1.2.14. AT+QCFG="risignaltype"	Set MAIN_RI Signal Output Carrier	44
2.1.2.15. AT+QCFG="setsyscfg"	Set Value(s) to System Configuration Files.....	45
2.1.2.16. AT+QCFG="simcfg"	Configure SIM	46
2.1.2.17. AT+QCFG="simcmd"	Switch Between SIMs	46
2.1.2.18. AT+QCFG="sim_status_check"	SIM Status Check	47
2.1.2.19. AT+QCFG="stored/cell"	Delete Stored Cell Acquisition DB	48
2.1.2.20. AT+QCFG="uartmapping"	Show Port and Function Mapping	49
2.1.2.21. AT+QCFG="uartmapping/profile"	User Port Mapping Profile	50
2.1.2.22. AT+QCFG="urc/delay"	Set URC Output Time	51
2.1.2.23. AT+QCFG="urc/ri/other"	Set MAIN_RI Behavior for Other URCs	52
2.1.2.24. AT+QCFG="urc/ri/ring"	Set MAIN_RI Behavior for RING URC	53
2.1.2.25. AT+QCFG="urc/ri/smsincoming"	Set MAIN_RI Behavior for Incoming SMS URCs	54
2.1.2.26. AT+QCFG="usb"	Activate/Deactivate USB Interface	55
2.1.2.27. AT+QCFG="usb/urcport"	Set URC Output Port to USB Interface Mode	56
2.1.2.28. AT+QCFG="cmux/urcport"	Set URC Output Port to CMUX Mode	57
2.1.2.29. AT+QCFG="fast/poweroff"	Control Fast Shutdown Function	58
2.1.2.30. AT+QCFG="dhclkrefresh"	Turn Off PMU LP Refresh Based on Current Temperature....	59
3	Summary of +CME ERROR Codes.....	60
4	Appendix References	62

Table Index

Table 1: Types of AT Commands	9
Table 2: Summary of +CME ERROR Codes	60
Table 3: Related Document.....	62
Table 4: Terms and Abbreviations	62

1 Introduction

This document describes the **AT+QCFG** commands supported on Quectel BG770A-GL, BG772A-GL, BG773A-GL, BG950A-GL, BG951A-GL, BG952A-GL, BG953A-GL, BG955A-GL and BG950S-GL modules.

1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

AT+QCFG commands implemented by the module are in "Extended" syntax, as illustrated below.

- **Extended**

Extended commands can be executed in several modes, as shown in the following table:

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

Multiple commands can be placed on a single line using a semi-colon (;) between commands. In such cases, only the first command should have **AT** prefix. Commands can be in upper or lower case.

Spaces should be ignored when you enter AT commands, except in the following cases:

- Within quoted strings, where spaces are preserved;
- Within an unquoted string or a numeric parameter;
- Within an IP address;
- Within the AT command name up to and including a =, ? or =?.

On input, at least a carriage return is required. A newline character is ignored so it is permissible to use carriage return/line feed pairs on the input.

If no command is entered after the **AT** token, **OK** will be returned. If an invalid command is entered, **ERROR** will be returned.

Optional parameters, unless explicitly stated, need to be provided up to the last entered parameter.

1.3. AT Command Responses

When the AT command processor has finished processing a line, it will output **OK**, **ERROR** or **+CME ERROR: <err>** to indicate that it is ready to accept a new command. Solicited information responses are sent before the final **OK**, **ERROR** or **+CME ERROR: <err>**.

Responses will be in the format of:

```
<CR><LF>+CMD1:<parameters><CR><LF>
<CR><LF>OK<CR><LF>
```

1.4. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence. The URLs, domain names, IP addresses, usernames/accounts, and passwords (if any) in the AT command examples are provided for illustrative and explanatory purposes only, and they should be modified to reflect your actual usage and specific needs.

2 Description of AT+QCFG Commands

2.1. AT+QCFG Extended Configuration Settings

The following Test Command shows the supported extended configuration settings of the module.

AT+QCFG Extended Configuration Settings

Test Command	Response
AT+QCFG=?	+QCFG: "airplanecontrol",(list of supported <airplane_control>s) +QCFG: "apn/display",(list of supported <mode>s) +QCFG: "apready",(list of supported <n>s),(list of supported <level>s),(range of supported <internal>s) +QCFG: "band",(range of supported <GSM_bandval>s),(range of supported <eMTC_bandval>s),(range of supported <NB-IoT_bandval>s),(list of supported <effect>s) +QCFG: "bandrestore" +QCFG: "celevel" +QCFG: "edrxusimact",(range of supported <mode>s) +QCFG: "dhclkrefresh",(list of supported <enable>s),(range of supported <upper_temp_limit>s) +QCFG: "emux/urcport",(range of supported <URC_port>s) +QCFG: "epco",(range of supported <mode>s) +QCFG: "gpio",(range of supported <mode>s),(list of supported <pin>s),(list of supported <dir>s),(range of supported <pull>s),(list of supported <drv>s),(list of supported <val>s),(list of supported <save>s) +QCFG: "nwscanmode",(list of supported <scan_mode>s),(list of supported <effect>s) +QCFG: "iotopmode",(range of supported <mode>s),(list of supported <effect>s) +QCFG: "lapiconf",<mode>,(list of supported <enable>s) +QCFG: "lte/bandprior",(range of supported <band_value>s) +QCFG: "autopwrkey",(range of supported <value>s) +QCFG: "automatic_recovery",(list of supported <enable>s) +QCFG: "lwm2m",(list of supported <enable>s) +QCFG: "netupd",(list of supported <cmd>s),(list of supported <mode>s) +QCFG: "nwoper",(list of supported <oper_name>s)

	+QCFG: "nwscanseq", (range of supported <scanseq>s), (list of supported <effect>s) +QCFG: "ntpsync", (list of supported <type>s), (list of supported <serveraddress>s)] +QCFG: "ps_dev_mob_type", (list of supported <flag>s) +QCFG: "extmobility", (range of supported <value>s) +QCFG: "psm/enter", (list of supported <mode>s) +QCFG: "psm/urc", (list of supported <enable>s) +QCFG: "risignaltype", (list of supported <RI_signal_type>s) +QCFG: "servicedomain", (list of supported <service>s) +QCFG: "setsyscfg", (list of supported <obj>s), (range of supported <stage>s), (list of supported <enable>s), (range of supported <lower_fr>s), (range of supported <upper_fr>s), (range of supported <state>s) +QCFG: "simcfg", <mode>, (list of supported <value>s) +QCFG: "simcmd", <mode>, (list of supported <value>s), (list of supported <value>s) +QCFG: "sim_status_check" +QCFG: "sibinfo" +QCFG: "emmtimer" +QCFG: "emmcause", (range of supported <display_format>s) +QCFG: "raiflag", (range of supported <value>s) +QCFG: "stored/cell", (list of supported <mode>s) +QCFG: "uartmapping" +QCFG: "uartmapping/profile", (list of supported <profile>s) +QCFG: "urc/delay", (list of supported <enable>s) +QCFG: "urc/ri/other", (list of supported <typeRI>s), (range of supported <pulse_duration>s), (range of supported <pulse_count>s) +QCFG: "urc/ri/ring", (list of supported <typeRI>s), (range of supported <pulse_duration>s), (range of supported <active_duration>s), (range of supported <inactive_duration>s), (list of supported <ring_no_disturbing>s), (range of supported <pulse_count>s) +QCFG: "urc/ri/smsincoming", (list of supported <typeRI>s), (range of supported <pulse_duration>s), (range of supported <pulse_count>s) +QCFG: "usb", (list of supported <mode>s) +QCFG: "usb/urcport", (range of supported <URC_port>s) +QCFG: "autoconnect", (range of supported <value>s) +QCFG: "cmux/urcport", (range of supported <URC_port>s) +QCFG: "fast/poweroff", (range of supported <value>s), (list of supported <enable>s) OK
Maximum Response Time	300 ms

NOTE

The response lines for **AT+QCFG=?** can be slightly different because specific commands are not supported by certain modules.

1. **AT+QCFG="nwscanmode"** is valid only on BG955A-GL.
2. **AT+QCFG="simcfg"**, **AT+QCFG="simcmd"** and **AT+QCFG="sim_status_check"** are valid only on BG953A-GL and BG773A-GL.

2.1.1. Network-Related AT Commands

NOTE

For BG950S-GL, all network-related AT commands except **AT+QCFG="iotopmode"** are in development.

2.1.1.1. AT+QCFG="band" Set Frequency Band

The command queries and sets the frequency bands to be searched.

AT+QCFG="band" Set Frequency Band

Write Command

AT+QCFG="band"[,<GSM_bandval>,<eMTC_bandval>,<NB-IoT_bandval>[,<effect>]]

Response

If the optional parameters are omitted, query the current setting:

+QCFG: "band",<GSM_bandval>,<eMTC_bandval>,<NB-IoT_bandval>

OK

If any of the optional parameters is specified, set the frequency bands to be searched.

OK

If there is an error related to ME functionality:

+CME ERROR: <err>

If there is any other error:

ERROR

Maximum Response Time

4500 ms

Characteristics

<effect> determines when the command takes effect.
The configurations are saved automatically.

Parameter

<GSM_bandval>	A hexadecimal value that specifies the GSM frequency band (e.g.: 0xa = 0x2(DCS1800) + 0x8(PCS1900)). If it is set to 0, it means not to change GSM frequency band. By default, all bands are set.
0	No change
0x1	EGSM900
0x2	DCS1800
0x4	GSM850
0x8	PCS1900
0xf	All of the supported bands above

<eMTC_bandval>	A hexadecimal value that specifies the eMTC frequency band (e.g.: 0x15 = 0x01(LTE B1) + 0x04(LTE B3) + 0x10(LTE B5)). If it is set to 0, the eMTC frequency band will not be changed. By default, all bands are set.
-----------------------------	--

0	No change
0x1 (BAND_PREF_LTE_BAND1)	LTE B1
0x2 (BAND_PREF_LTE_BAND2)	LTE B2
0x4 (BAND_PREF_LTE_BAND3)	LTE B3
0x8 (BAND_PREF_LTE_BAND4)	LTE B4
0x10 (BAND_PREF_LTE_BAND5)	LTE B5
0x80 (BAND_PREF_LTE_BAND8)	LTE B8
0x800 (BAND_PREF_LTE_BAND12)	LTE B12
0x1000 (BAND_PREF_LTE_BAND13)	LTE B13
0x20000 (BAND_PREF_LTE_BAND18)	LTE B18
0x40000 (BAND_PREF_LTE_BAND19)	LTE B19
0x80000 (BAND_PREF_LTE_BAND20)	LTE B20
0x1000000 (BAND_PREF_LTE_BAND25)	LTE B25
0x2000000 (BAND_PREF_LTE_BAND26)	LTE B26
0x4000000 (BAND_PREF_LTE_BAND27)	LTE B27
0x8000000 (BAND_PREF_LTE_BAND28)	LTE B28
0x2000000000000000 (BAND_PREF_LTE_BAND66)	LTE B66

<NB-IoT_bandval>	A hexadecimal value that specifies the NB-IoT frequency band (e.g.: 0x15 = 0x01(LTE B1) + 0x04(LTE B3) + 0x10(LTE B5)). If it is set to 0, the NB-IoT frequency band will not be changed. By default, all bands are set.
-------------------------------	--

0	No change
0x1 (BAND_PREF_LTE_BAND1)	LTE B1
0x2 (BAND_PREF_LTE_BAND2)	LTE B2
0x4 (BAND_PREF_LTE_BAND3)	LTE B3
0x8 (BAND_PREF_LTE_BAND4)	LTE B4
0x10 (BAND_PREF_LTE_BAND5)	LTE B5
0x80 (BAND_PREF_LTE_BAND8)	LTE B8
0x800 (BAND_PREF_LTE_BAND12)	LTE B12
0x1000 (BAND_PREF_LTE_BAND13)	LTE B13
0x10000 (BAND_PREF_LTE_BAND17)	LTE B17

	0x20000 (BAND_PREF_LTE_BAND18)	LTE B18
	0x40000 (BAND_PREF_LTE_BAND19)	LTE B19
	0x80000 (BAND_PREF_LTE_BAND20)	LTE B20
	0x1000000 (BAND_PREF_LTE_BAND25)	LTE B25
	0x8000000 (BAND_PREF_LTE_BAND28)	LTE B28
	0x2000000000000000 (BAND_PREF_LTE_BAND66)	LTE B66
<effect>	Integer type. Determines when the command takes effect.	
	0	Take effect after the module reboots
	1	Take effect immediately
<err>	Integer type. Error code. See Chapter 3 for more information.	

NOTE

<GSM_bandval> is valid only on BG955A-GL.

2.1.1.2. AT+QCFG="bandrestore" Restore Default Band Configuration

The Write Command restores the default band configuration.

AT+QCFG="bandrestore" Restore Default Band Configuration	
Write Command AT+QCFG="bandrestore"	Response OK
	If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots.

2.1.1.3. AT+QCFG="celevel" Query NB-IoT Coverage Enhancement Level

The command queries the NB-IoT coverage enhancement (CE) level.

AT+QCFG="celevel" Query NB-IoT Coverage Enhancement Level	
Write Command AT+QCFG="celevel"	Response: +QCFG: "celevel",<level>
	OK
	If there is an error related to ME functionality:

	+CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<level>	Integer type. NB-IoT coverage enhancement level. 0 CE level 0 1 CE level 1 2 CE level 2
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.4. AT+QCFG="epco" Set PCO Encoding Type: PCO or ePCO

The command queries and sets the type of PCO.

AT+QCFG="epco" Set PCO Encoding Type: PCO or ePCO	
Write Command AT+QCFG="epco"[,<value>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "epco"[,<value>] OK If the optional parameter is specified, set PCO encoding type: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<value>	Integer type. <u>0</u> Force UE (in NB-IoT or CAT M mode) to use PCO 1 Force UE (in NB-IoT or CAT M mode) to use ePCO
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.5. AT+QCFG="nwscanmode" Set RAT(s) to be Searched for

The command queries and sets the RAT(s) to be searched for.

AT+QCFG="nwscanmode" Set RAT(s) to be Searched for

Write Command AT+QCFG="nwscanmode"[,<scan_mode>,<effect>]	Response If the optional parameters are omitted, query the current setting: +QCFG: "nwscanmode",<scan_mode> OK If any of the optional parameter is specified, set the RAT(s) to be searched for: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	<effect> determines when the command takes effect. The configurations are saved automatically.

Parameter

<scan_mode>	Integer type. RAT(s) to be searched for. <u>0</u> Automatic (GSM and LTE) 1 GSM only 3 LTE only
<effect>	Integer type. When to take effect. 0 Take effect after rebooting <u>1</u> Take effect immediately
<err>	Error code. See Chapter 3 for more information.

NOTE

This command is valid only on BG955A-GL.

2.1.1.6. AT+QCFG="iotopmode" Set Network Category to be Searched Under LTE RAT

The command queries and sets the network category to be searched under LTE RAT.

AT+QCFG="iotopmode" Set Network Category to be Searched Under LTE RAT

Write Command AT+QCFG="iotopmode"[,<mode>[,<effect>]]	Response If the optional parameters are omitted, query the current setting: +QCFG: "iotopmode",<mode> OK If any of the optional parameters is specified, set the network category to be searched under LTE RAT: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	4500 ms
Characteristics	<effect> determines when the command takes effect. The configurations are saved automatically.

Parameter

<mode>	Integer type. Network category to be searched under LTE RAT. 0 eMTC 1 NB-IoT <u>2</u> eMTC and NB-IoT
<effect>	Integer type. Determines when the command takes effect. 0 Take effect after the module reboots <u>1</u> Take effect immediately
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.7. AT+QCFG="lapiconf" Set LAPI Value (Low Access Priority Indicator)

The command queries the current settings or sets the Low Access Priority Indicator.

AT+QCFG="lapiconf" Set LAPI Value (Low Access Priority Indicator)

Write Command

AT+QCFG="lapiconf"[,<mode>][,<enable>]

Response

If the optional parameters are omitted, query the current setting:

+QCFG: "lapiconf",<mode>[,<enable>]

OK

If any of the optional parameters is specified, set the LAPI value:

OK

If there is an error related to ME functionality:

+CME ERROR: <err>

If there is any other error:

ERROR

Maximum Response Time

300 ms

Characteristics

The configuration is saved automatically.

Parameter

<mode>	Integer type. Whether to enable low access priority indication feature 0 Disable. Forced to disable 1 Enable. Forced to enable 2 Auto. Determined by EFS file
<enable>	Integer type. 0 Disable 1 Enable
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.8. AT+QCFG="nwscanseq" Set RATs Searching Sequence

The command queries and sets RATs searching sequence.

AT+QCFG="nwscanseq" Set RATs Searching Sequence

Write Command

AT+QCFG="nwscanseq"[,<scanseq>[,<effect>]]

Response

If the optional parameters are omitted, query the current setting:

+QCFG: "nwscanseq",<scanseq>

OK

If any of the optional parameters is specified, set RATs searching sequence:

OK

If there is an error related to ME functionality:

+CME ERROR: <err>

If there is any other error:

ERROR

Maximum Response Time

300 ms

Characteristics

<effect> determines when the command takes effect.
The configurations are saved automatically.

Parameter

<scanseq> Integer type. RATs searching sequence. (e.g.: 020301 stands for eMTC → NB-IoT → GSM)

00 Automatic (eMTC → NB-IoT)

01 GSM

02 eMTC

03 NB-IoT

<effect> Integer type. Determines when the command takes effect.

0 Take effect after the module reboots

1 Take effect immediately

<err> Integer type. Error code. See **Chapter 3** for more information.

2.1.1.9. AT+QCFG="ntpsync" Synchronize Time in Milliseconds with NTP Server

This command synchronizes the local time accuracy in milliseconds with the Universal Time Coordinated (UTC) through the NTP server. This local time in milliseconds can be queried by **AT+QCCLK** command. For more details of **AT+QCCLK**, see *document [1]*.

AT+QCFG="ntpsync" Synchronize Time in Milliseconds with NTP Server

Write Command AT+QCFG="ntpsync",<type>[,<serveraddress>]	Response OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<type>	Integer type. Operating mode. 1 Trigger Time Synchronization
<serveraddress>	String type. NTP server address.

NOTE

This command only works with **AT+QCCLK**. It should be triggered before **AT+QCCLK** is used.

2.1.1.10. AT+QCFG="extmobility" Set Mobility Type

The command queries and sets mobility type.

AT+QCFG="extmobility" Set Mobility Type

Write Command AT+QCFG="extmobility"[,<value>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "extmobility"[,<value>] OK If the optional parameter is specified, set mobility type: OK If there is an error related to ME functionality:
---	--

	+CME ERROR: <err>
	If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<value>	Integer type. Mobility type. <u>0</u> SW default 1 Mobile 2 Static 3 Mobile power optimized 4 Static power optimized
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.11. AT+QCFG="psm/enter" Trigger Module into PSM Immediately

The command queries the current settings or sets whether to trigger the module into the Power Saving Mode (PSM) immediately after the RRC connection release is received.

When **<mode>=1**, the module skips the active timer (T3324) and enters PSM immediately after the RRC connection release is received.

AT+QCFG="psm/enter" Trigger Module into PSM Immediately

Write Command AT+QCFG="psm/enter"[,<mode>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "psm/enter",<mode> OK If the optional parameter is specified, set whether to trigger the module into PSM immediately after the RRC connection release is received: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error:
--	---

	ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<mode>	Integer type. Sets whether to trigger the module into PSM. 0 Enter PSM after T3324 expires. 1 Enter PSM immediately after the RRC connection release is received.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.12. AT+QCFG="psm/urc" Enable/Disable PSM Entering Indication

The command queries the current settings and enables/disables the output of URC **+QPSMTIMER: <TAU_timer>,<T3324_timer>**, which indicates the TAU duration and active time duration of the module's PSM. For more details of this URC, see **document [2]**.

When the PSM function is enabled and the RRC connection release is received, the active timer (T3324) is started and the indication URC is reported.

AT+QCFG="psm/urc" Enable/Disable PSM Entering Indication

Write Command AT+QCFG="psm/urc"[,<enable>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "psm/urc",<enable> OK If the optional parameter is specified, set whether to enable PSM entering indication. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<enable>	Integer type. Enable/disable the output of URC +QPSMTIMER: <TAU_timer>, <T3324_timer> . If enabled, the URC is reported when the RRC connection release is received. For more details of this URC, see document [2] . 0 Disable 1 Enable
<TAU_timer>	Integer type. The TAU duration of PSM. Unit: second.
<T3324_timer>	Integer type. The active time duration of PSM. Unit: second.
<err>	Integer type. Error code. See Chapter 3 for more information.

NOTE

When **AT+QCFG="psm/urc",1** and **AT+QCFG="psm/enter",1** are executed at the same time, it is possible that the URC **+QPSMTIMER: <TAU_timer>, <T3324_timer>** cannot be outputted because the module enters PSM immediately. For more details of this URC, see **document [2]**.

2.1.1.13. AT+QCFG="servicedomain" Set Service Domain

The command queries and sets the service domain to be registered.

AT+QCFG="servicedomain" Set Service Domain

Write Command AT+QCFG="servicedomain"[,<service>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "servicedomain",<service> OK If the optional parameter is specified, set the service domain to be registered: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The configurations are saved automatically.

Parameter

<service>	Integer type. Service domain. 1 PS only
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.14. AT+QCFG="sibinfo" Query SIB Information

The Command queries the SIB information.

AT+QCFG="sibinfo" Query SIB Information	
Write Command AT+QCFG="sibinfo"	Response +QCFG: "sibinfo",<EARFCN>,<PCI>,<q_RX_lev_min>,[<q_qual_min>],<s_intra_search>,<cell_resel_priority>,<thresh_serving_low>,[<s_non_intra_search>],[<idle_DRX_cycle_len>] OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	/

Parameter

<EARFCN>	Integer type. EARFCN of the serving cell. Range: 0-65535.
<PCI>	Integer type. Physical cell ID. Range: 0-503.
<q_RX_lev_min>	Integer type. Minimum required RX level in the cell. Range: 0-140. Unit: dBm.
<q_qual_min>	Integer type. Minimum required quality level in the cell. Range: 0-140. Unit: dB.
<s_intra_search>	Integer type. Cell selection parameter that specifies the Srxlev threshold for intra-frequency measurements. Unit: dB.
<cell_resel_priority>	Integer type. Cell reselection priority. Range: 0-7. Value 0 means lowest priority.
<thresh_serving_low>	Integer type. Specifies the suitable reception level threshold used by the UE. On the serving cell when reselecting towards a lower-priority RAT/frequency. Unit: dB.
<s_non_intra_search>	Integer type. Threshold to control non-intra-frequency searches.

<idle_DRX_cycle_len>	Integer type. Idle DRX cycle length. Unit: ms.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.15. AT+QCFG="emmtimer" Query EMM Timer

The command queries EPS mobility management timer.

AT+QCFG="emmtimer" Query EMM Timer

Write Command AT+QCFG="emmtimer"	Response +QCFG: "emmtimer",<T3402_value>,<T3412_value> OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<T3402_value>	Integer type. T3402 timer value. Default: 0. Unit: second.
<T3412_value>	Integer type. T3412 timer value. Default: 6840. Unit: second.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.16. AT+QCFG="emmcause" Get EMM Cause Value

This Write Command queries the EMM cause value for the rejected attach request.

AT+QCFG="emmcause" Get EMM Cause Value

Write Command AT+QCFG="emmcause"[,<display_format>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "emmcause"[,<cause_value>] OK If the optional parameter is specified, set display format: OK
---	---

	If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<display_format>	Integer type. Display format of EMM cause value. 0 Numeric value 1 Verbose value
<cause_value>	EMM reject cause value. See <i>3GPP 24.301</i> for details.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.1.17. AT+QCFG="raiflag" Set RAI Flag

The command queries and sets RAI (Release Assistance Indication) flag.

AT+QCFG="raiflag" Set RAI Flag	
Write Command AT+QCFG="raiflag"[,<value>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "raiflag"[,<value>] OK If the optional parameter is specified, set the RAI flag: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<value>	Integer type. RAI settings: 0 No RAI flag 1 Last UL packet is expected 2 Last UL and single DL packets are expected
<err>	Integer type. Error code. See Chapter 3 for more information.

NOTE

- RAI flag is only supported for UDP transfers and for firmware version R02. R02 firmware refers to the firmware whose version number contains a segment "R02", such as "BG770AGL...R02A01...".
- After setting RAI flag (**AT+QCFG="raiflag",1** or **AT+QCFG="raiflag",2**), if the last packet is not sent, it will be automatically cleared after certain interval, which is predefined by the base station.

Example

```

AT+QIOPEN=1,0,"UDP","93.86.51.214",35750,35750,1 //Open a UDP socket
OK
+QIOPEN: 0,0
AT+QCSCON=1 //Enable reporting URC about connection
status
OK
AT+QCFG="raiflag",1 //Set RAI flag to 1 (release immediately
after a packet is sent)
OK
AT+QISEND=0
> quectel
SEND OK

+QCSCON: 1,1
+QCSCON: 1,0 //RRC connection is immediately
released, and the module receives a
download packet after that

+QCSCON: 1,1
+QIURC: "recv",0,7
quectel
+QCSCON: 1,0

AT+QCFG="raiflag",2 //Set RAI flag to 2 (don't release before
the download packet is received)

```

```

OK
AT+QISEND=0
> quectel
SEND OK

+QCSCON: 1,1
+QIURC: "recv",0,7
quectel
+QCSCON: 1,0                                     //RRC connection is immediately released
                                                    and the module receives a download packet
                                                    after that

AT+QICLOSE=0                                     //Close the UDP socket
OK
    
```

2.1.1.18. AT+QCFG="autoconnect" Enable/Disable Auto Connect Mode

The command queries and sets auto connect mode.

AT+QCFG="autoconnect" Enable/Disable Auto Connect Mode

Write Command	Response
AT+QCFG="autoconnect"[,<value>]	If the optional parameter is omitted, query the current setting: +QCFG: "autoconnect"[,<value>]
	OK
	If the optional parameter is specified, set auto connect mode: OK
	If there is an error related to ME functionality: +CME ERROR: <err>
	If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately.

Parameter

<value>	Integer type. Whether to enable/disable auto connect mode.
0	Disable auto connect
<u>1</u>	Enable auto connect

<err> Integer type. Error code. See **Chapter 3** for more information.

2.1.2. Platform-Related AT Commands

Among the following AT commands, **AT+QCFG="urc/ri/ring"**, **AT+QCFG="urc/ri/smsincoming"** and **AT+QCFG="urc/ri/other"** commands control the behavior of the MAIN_RI pin when a URC is reported. The MAIN_RI pin is active low.

NOTE

For BG950S-GL, following commands are under development:

- **AT+QCFG="risignalttype"**
- **AT+QCFG="emux/urcport"**
- **AT+QCFG="apready"**
- **AT+QCFG="apn/display"**
- **AT+QCFG="lwm2m"**
- **AT+QCFG="nwoper"**
- **AT+QCFG="ps_dev_mob_type"**
- **AT+QCFG="netupd"**
- **AT+QCFG="edrxusimact"**
- **AT+QCFG="uartmapping"**
- **AT+QCFG="uartmapping/profile"**
- **AT+QCFG="stored/cell"**

2.1.2.1. AT+QCFG="airplanecontrol" Enable/Disable Airplane Mode Control via W_DISABLE#

The command enables/disables airplane mode control via the W_DISABLE# pin and queries the current settings. If the function is enabled, the module enters the airplane mode when the pin is pulled down, or the normal mode when the pin is pulled up. In addition, the command outputs a related URC before the module enters or exits airplane mode.

AT+QCFG="airplanecontrol" Enable/Disable Airplane Mode Control via W_DISABLE#

Write Command

AT+QCFG="airplanecontrol"[,<airplane_control>]

Response

If the optional parameter is omitted, query the current setting:

+QCFG: "airplanecontrol",<airplane_control>,<airplane_status>

OK

	If the optional parameter is specified, set whether to enable airplane mode control via W_DISABLE# pin: OK Or ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<airplane_control>	Integer type. Enable/disable airplane mode control via W_DISABLE# pin. 0 Disable 1 Enable
<airplane_status>	Integer type. Current status of the module. 0 In normal mode 1 In airplane mode

NOTE

The status of the W_DISABLE# pin may affect the validity of **AT+CFUN** (See **document [1]** for more information). When the airplane mode control via W_DISABLE# is enabled and the pin is pulled down, the module enters the airplane mode regardless of the module status set by **AT+CFUN**. In addition, the module functionality level cannot be switched with **AT+CFUN**.

Example

```

AT+QCFG="airplanecontrol",1 //Enable airplane mode control via W_DISABLE#.
OK
//Pull down W_DISABLE# pin.
+QIND: "airplanestatus",1 //URC indicating that the module entered airplane mode.

AT+QCFG="airplanecontrol" //Query whether airplane mode control via W_DISABLE# is enabled
and the current status of the module.
+QCFG: "airplanecontrol",1,1 //Airplane mode control via W_DISABLE# is enabled and the module
is currently in airplane mode.

OK
//Pull up W_DISABLE# pin
+QIND: "airplanestatus",0 //URC indicating that the module exited airplane mode.

```



```

AT+QCFG="airplanecontrol" //Query whether airplane mode control via W_DISABLE# is enabled
                           and the current status of the module.
+QCFG: "airplanecontrol",1,0 //Airplane mode control via W_DISABLE# is enabled and the module
                             is currently in normal mode.

OK
    
```

2.1.2.2. AT+QCFG="apn/display" Enable/Disable Showing APN

The command enables/disables showing of allocated point name (APN) and queries the current settings. When **<mode>**=1, the APN is shown in the response of **AT+CGDCONT?** (See **document [1]** for more information).

AT+QCFG="apn/display" Enable/Disable Showing APN

Write Command AT+QCFG="apn/display"[,<mode>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "apn/display",<mode> OK If the optional parameter is specified, set whether to enable showing APN. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<mode>	Integer type. Enable/disable showing APN. 0 Disable 1 Enable
<err>	Integer type. Error code. Refer to Chapter 3 for more information.

2.1.1.2.3. AT+QCFG="apready" Set AP_READY Behavior

The command queries and sets the behavior of the AP_READY pin. An external MCU can change AP_READY pin level as needed.

When there is a URC to be reported, if the AP_READY pin level is invalid, the URC is buffered first, and the AP_READY pin level will be detected periodically within the configured detection period. The URC will be output when the AP_READY pin level becomes valid. The pulse signal generated on the MAIN_RI pin can still be output according to the configured mode, and will not be buffered.

AT+QCFG="apready" Set AP_READY Behavior

Write Command AT+QCFG="apready"[,<n>[,<level>[,<interval>]]]	Response If the optional parameters are omitted, query the current setting: +QCFG: "apready",<n>,<level>,<interval> OK If any of the optional parameters is specified, set the AP_READY behavior: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<n>	Integer type. Enable/disable AP_READY pin for AP sleep state detection. 0 Disable 1 Enable
<level>	Integer type. Valid level of AP_READY pin. Parameter is valid only when AP_READY pin detection function is enabled. 0 Low level 1 High level
<interval>	Integer type. Detection interval. Range: 100–3000. Default: 500. Unit: ms. Parameter is valid only when the AP_READY detection function is enabled.

<err> Integer type. Error code. See **Chapter 3** for more information.

NOTE

1. Up to 15 URCs can be buffered. When the number of URCs exceeds 15, the oldest one in the buffer will be cleared to store a new URC.
2. **RING** URC is buffered only once for each call process.

2.1.2.4. AT+QCFG="automatic_recovery" Enable/Disable Automatic Recovery

The command enables/disables automatic recovery.

AT+QCFG="automatic_recovery" Enable/Disable Automatic Recovery

Write Command AT+QCFG="automatic_recovery"[,<enable>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "automatic_recovery",<enable> OK If the optional parameter is specified, enable or disable automatic recovery: OK Or ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<enable>	Integer type. Enable automatic recovery.
0	Disable automatic recovery
1	Enable automatic recovery

2.1.2.5. AT+QCFG="edrxusimact" Activate/Deactivate (U)SIM in eDRX Mode

The command activates/deactivates (U)SIM in eDRX mode regardless of (U)SIM EF_AD (Elementary File Access Data) settings.

AT+QCFG="edrxusimact" Activate/Deactivate (U)SIM in eDRX Mode

Write Command AT+QCFG="edrxusimact"[,<mode>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "edrxusimact",<mode> OK If the optional parameter is specified, activate/deactivate (U)SIM in eDRX mode regardless of (U)SIM EF_AD settings: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is not saved.

Parameter

<mode>	Integer type. (U)SIM activation policy. <u>0</u> Apply default (U)SIM activation policy as per (U)SIM EF_AD settings in eDRX mode 1 Activate (U)SIM in eDRX mode regardless of (U)SIM EF_AD settings 2 Deactivate (U)SIM in eDRX mode regardless of (U)SIM EF_AD settings
<err>	Integer type. Error code. See Chapter 3 for more information.

Example

```

AT+QCFG="edrxusimact"           //Query the current (U)SIM activation policy.
+QCFG: "edrxusimact",0

OK

```

2.1.2.6. AT+QCFG="emux/urcport" Set URC Output Port to EMUX Mode

The command queries and sets the URC output port to EMUX mode.

AT+QCFG="emux/urcport" Set URC Output Port to EMUX Mode

Write Command AT+QCFG="emux/urcport"[,<URC_port>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "emux/urcport",<URC_port> OK If the optional parameter is specified, set the URC output port to EMUX mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<URC_port>	Integer type. URC output port in EMUX mode. 0 All ports 1 Virtual port 1 2 Virtual port 2 3 Virtual port 3
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.7. AT+QCFG="gpio" Set GPIO Status

The command queries and sets the GPIO status.

AT+QCFG="gpio" Set GPIO Status

Write Command Query the formats of the command AT+QCFG="gpio"	Response +QCFG: "gpio",<mode>,<pin>[,<dir>,<pull>,<drv>][<val>][,<save>]]
--	---

	OK
Write Command AT+QCFG="gpio",<mode>,<pin>[,<dir>,<pull>,<drv>][,<val>][,<save>]]	Response If <mode>=2, omit all optional parameters. +QCFG: "gpio",<val> OK If <mode>=1, omit <val>. OK If <mode>=3, omit <dir>, <pull> and <drv>. OK If there is an error related to ME functionality: +CME ERROR: <err>
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. <save> determines whether the configuration will be saved. When <save>=1, the configurations are saved and remain valid after the module reboots.

Parameter

<mode>	Integer type. Command mode. 1 Initialize GPIO status 2 Query GPIO status 3 Set GPIO status																						
<pin>	Integer type. GPIO pin number. BG770A-GL and BG773A-GL modules support the following pin numbers: <table> <tr> <th>Pin No.</th><th>Pin Name</th></tr> <tr> <td>1</td><td>GPIO1</td></tr> <tr> <td>8</td><td>GPIO2</td></tr> <tr> <td>9</td><td>GPIO3</td></tr> <tr> <td>33</td><td>GPIO4</td></tr> <tr> <td>40</td><td>GPIO5</td></tr> <tr> <td>57</td><td>GPIO6</td></tr> <tr> <td>63</td><td>GPIO7</td></tr> </table> BG950A-GL, BG951A-GL, BG953A-GL, BG950S-GL and BG955A-GL modules support the following pin numbers: <table> <tr> <th>Pin No.</th><th>Pin Name</th></tr> <tr> <td>25</td><td>GPIO1</td></tr> <tr> <td>26</td><td>GPIO2</td></tr> </table>	Pin No.	Pin Name	1	GPIO1	8	GPIO2	9	GPIO3	33	GPIO4	40	GPIO5	57	GPIO6	63	GPIO7	Pin No.	Pin Name	25	GPIO1	26	GPIO2
Pin No.	Pin Name																						
1	GPIO1																						
8	GPIO2																						
9	GPIO3																						
33	GPIO4																						
40	GPIO5																						
57	GPIO6																						
63	GPIO7																						
Pin No.	Pin Name																						
25	GPIO1																						
26	GPIO2																						

	64	GPIO3
	65	GPIO4
	66	GPIO5
	85	GPIO6
	86	GPIO7
	87	GPIO8
	88	GPIO9
<dir>	Integer type. GPIO pin direction.	
	0	Input
	1	Output
<pull>	Integer type. GPIO pin pull type.	
	0	No pull
	1	Pull the GPIO down
	2	Keep the GPIO as it is
	3	Pull the GPIO up
<drv>	Integer type. GPIO pin drive strength.	
	0	2 mA
	1	4 mA
	3	8 mA
	5	12 mA
<val>	Integer type. Value read from or written to a GPIO.	
	0	Low level
	1	High level
<save>	Integer type. Determines whether to save the configurations.	
	0	Do not save
	1	Save
<err>	Integer type. Error code. See Chapter 3 for more information.	

NOTE

1. Command **AT+QCFG="gpio"** cannot be used on BG772A-GL and BG952A-GL modules.
2. **<save>** is valid only when **<mode>** is 1 or 3.
3. For the value of **<pin>**, see corresponding hardware design for details.

2.1.2.8. AT+QCFG="lte/bandprior" Set Band Scan Priority Under LTE

The command queries and sets the band scan priority under LTE.

AT+QCFG="lte/bandprior" Set Band Scan Priority Under LTE

Write Command	Response
AT+QCFG="lte/bandprior" [<band_value>]	If the optional parameter is omitted, query the current setting: +QCFG: "lte/bandprior",<band_value>

	OK If the optional parameter is specified, set band scan priority under LTE: OK If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The configuration is saved automatically.

Parameter

<band_value>	LTE Cat M1 band value
01	Band 1
02	Band 2
03	Band 3
04	Band 4
05	Band 5
07	Band 7
08	Band 8

Example

```

AT+QCFG="lte/bandprior",1,5,7 //Set Band 1, Band 5 and Band 7.
OK //Search priority: 1 -> 5 -> 7
AT+QCFG="lte/bandprior" //Query band scan priority under LTE
+QCFG="lte/bandprior",1,5,7
OK

```

2.1.2.9. AT+QCFG="autopwrkey" Enable/Disable Autoboot

The command enables/disables auto power-on procedure.

AT+QCFG="autopwrkey" Enable/Disable Autoboot

Write Command	Response
AT+QCFG="autopwrkey"[,<value>]	If the optional parameter is omitted, query the current setting: +QCFG: "autopwrkey",<value>
	OK

	If the optional parameter is specified, set whether to enable autoboot: OK If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The configuration is saved automatically.

Parameter

<value>	Integer value. <u>0</u> Enable auto power-on; Module will power on w/o using PWRKEY 1 Disable auto power-on; PWRKEY should be used to power on the module
----------------------	--

2.1.2.10. AT+QCFG="lwm2m" Enable/Disable LwM2M

The command enables/disables LwM2M and queries the current settings.

AT+QCFG="lwm2m" Enable/Disable LwM2M	
Write Command AT+QCFG="lwm2m",<enable>	Response If the optional parameter is omitted, query the current setting: +QCFG: "lwm2m",<enable> OK If the optional parameter is specified, set whether to enable LwM2M. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<enable>	Integer type. Whether to enable LwM2M. 0 Disable 1 Enable
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.11. AT+QCFG="netupd" Enable/Disable Network Override for Specified LTE Parameters

The command queries the current settings and enables/disables network override for specified LTE parameters (i.e., by EMM messages).

AT+QCFG="netupd" Enable/Disable Network Override for Specified LTE Parameters

Write Command AT+QCFG="netupd",<cmd>[,<mode>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "netupd",<cmd>,<mode> OK If the optional parameter is specified, enable/disable network override for specified LTE parameters: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are not saved.

Parameter

<cmd>	String type. "nwname" Set the behavior of network name supplied by AT+COPS? .
<mode>	Integer type. For "nwname": 0 AT+COPS? shows the most updated full network name as required by <i>3GPP TS 27.007</i> standard.

- 1 Prohibit override of network name by EMM message (i.e. shown in **AT+COPS?**, etc.) even if the EMM information message indicates another full network name.

<err> Integer type. Error code. See **Chapter 3** for more information.

Example

```
AT+QCFG="netupd","nwname",1 //Set "nwname" to mode 1.
OK
```

2.1.2.12. AT+QCFG="nwoper" Set Network Operator Mode of Modem

The command queries and sets network operator mode of the modem. This mode setting supports network operator specific requirements defined on top of 3GPP requirements.

AT+QCFG="nwoper" Set Network Operator Mode of Modem

Write Command AT+QCFG="nwoper"[,<oper_name>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "nwoper",<oper_name>[,<mode>] OK If the optional parameter is specified, set network operator mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<oper_name>	String type. Network operator mode of modem.
"DEFAULT"	Default 3GPP compliant mode
"VZW"	Verizon
"ATT"	AT&T
...	

<mode>	String type. Optional parameter indicates that "AUTO" mode is currently configured. "AUTO" Network operator mode is selected based on IMSI value of currently used (U)SIM
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.13. AT+QCFG="ps_dev_mob_type" Set Device Mobility Type Flag

The command queries and sets device mobility type flag. It gets the configuration from JSON configuration file.

AT+QCFG="ps_dev_mob_type" Set Device Mobility Type Flag

Write Command AT+QCFG="ps_dev_mob_type"[,<flag>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "ps_dev_mob_type",<flag> OK If the optional parameter is specified, set device mobility type flag: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<flag>	Integer type. Device mobility type flag. <u>1</u> Mobile 2 Static
<err>	Integer type. Error code. See Chapter 3 for more information.

Example

```
AT+QCFG="ps_dev_mob_type",1 //Set the flag to mobile.
OK
```

2.1.2.14. AT+QCFG="risignalttype" Set MAIN_RI Signal Output Carrier

The command queries and sets the output carrier of MAIN_RI signal.

AT+QCFG="risignalttype" Set Main_RI Signal Output Carrier	
Write Command AT+QCFG="risignalttype"[,<RI_signal_type>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "risignalttype",<RI_signal_type> OK If the optional parameter is specified, set the output carrier of the MAIN_RI signal. OK If there is an error related to the ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<RI_signal_type>	String type. MAIN_RI signal output carrier.
	<u>"respective"</u> Causes MAIN_RI to behave according to the port where a URC is reported. For example, if a URC is reported on the UART port, MAIN_RI acts as a physical ring indication signal. Use AT+QURCCFG="urcport" to determine the URC presentation port (See document [1] for more information). "physical" Regardless of URC presentation port, MAIN_RI behaves strictly as a physical ring indication signal.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.15. AT+QCFG="setsyscfg" Set Value(s) to System Configuration Files

The command sets value(s) to system configuration files.

AT+QCFG="setsyscfg" Set Value(s) to System Configuration Files

Write Command AT+QCFG="setsyscfg",<obj>,<stage>,<enable>,<lower_fr>,<upper_fr>,<state>	Response OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module reboots. The configurations are saved automatically.

Parameter

<obj>	String type. "ant_cfg" Antenna tuning configuration file.
<stage>	Integer type. Antenna characterization stage. Range: 1–56.
<enable>	Integer type. 0 Enable corresponding stage 1 Disable corresponding stage
<lower_fr>	Integer type. Lower frequency of corresponding bandwidth, given in MHz multiplied by 10. Range: 4000–38000.
<upper_fr>	Integer type. Upper frequency of corresponding bandwidth, given in MHz multiplied by 10. Range: 4000–38000.
<state>	Integer type. Truth table combination of two different states of GRFC1 and GRFC2 pins respectively. Range: 0–3.
<err>	Integer type. Error code. See Chapter 3 for more information.

Example

```
AT+QCFG="setsyscfg","ant_cfg",1,1,8031,9600,2 // 803.1-960MHz
OK
```

NOTE

Overlapping between frequency ranges is forbidden.

2.1.2.16. AT+QCFG="simcfg" Configure SIM

The command queries and configures SIM settings.

AT+QCFG="simcfg" Configure SIM

Write Command Set wakeup SIM selection policy AT+QCFG="simcfg","sim_init_select_policy",[,<sim_init_select_policy>]	Response If the optional parameter is omitted, query the current setting. +QCFG: "simcfg","sim_init_select_policy",<sim_init_select_policy> OK If the optional parameter is specified, set wakeup SIM selection policy. OK Or ERROR
Maximum Response Time	300 ms
Characteristics	-

Parameter

<sim_init_select_policy>	Integer type. SIM selection policy.
0	N/A, Single SIM
1	SIM only
5	IUICC (iSIM) only option (no fallback available)

NOTE

This command is valid only on BG773A-GL and BG953A-GL modules.

2.1.2.17. AT+QCFG="simcmd" Switch Between SIMs

The command switches between SIMs (USIM/eSIM and iSIM).

AT+QCFG="simcmd" Switch Between SIMs

Write Command AT+QCFG="simcmd",<cmd>,<selection>[,<persistence>]	Response OK If there is any error
--	--

	ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<cmd>	String type. Operation to be applied. "SWITCH" Switch to another SIM (USIM/eSIM or iSIM). The attempt to switch to the SIM already in use will be ignored and return OK . The attempt to switch to the SIM disabled in BSP file will return ERROR .
<selection>	Integer type. SIM selection policy to be applied immediately. 1 USIM/eSIM 3 iSIM (iUICC)
<persistence>	Integer type. Persistence of the setting. 0 Not persistent (By default if the parameter is omitted) 1 Persistent

NOTE

This command is valid only on BG773A-GL and BG953A-GL modules.

2.1.2.18. AT+QCFG="sim_status_check" SIM Status Check

This command checks which SIM (USIM/eSIM or iSIM) is in use.

AT+QCFG="sim_status_check" SIM Status Check

Write Command AT+QCFG="sim_status_check"	Response +QCFG: "sim_status_check",<status> OK
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<status>	Integer type. SIM in use.
0	SIM not selected
1	USIM/eSIM selected
3	iSIM (iUICC) selected

NOTE

This command is valid only on BG773A-GL and BG953A-GL modules.

2.1.2.19. AT+QCFG="stored/cell" Delete Stored Cell Acquisition DB

The command deletes stored cell acquisition DB (Data Base).

AT+QCFG="stored/cell" Delete Stored Cell Acquisition DB

Write Command AT+QCFG="stored/cell"[,<value>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "stored/cell",<value> OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The configuration is saved automatically.

Parameter

<value>	Integer value.
0	Delete acquisition DB under NB-IoT
1	Delete acquisition DB under eMTC

2.1.2.20. AT+QCFG="uartmapping" Show Port and Function Mapping

The command presents the mapping of ports (physical and virtual), i.e., to provide an overview of the functions performed by each port.

AT+QCFG="uartmapping" Show Port and Function Mapping

Write Command

AT+QCFG="uartmapping"

Response

<port>:<function>,

...

<port>:<function>

OK

If there is an error related to ME functionality:

+CME ERROR: <err>

If there is any other error:

ERROR

Maximum Response Time

500 ms

Characteristics

The command takes effect immediately.

Parameter

<port>

String type.

DBG

Physical port, DBG UART

MAIN

Physical port, MAIN UART

AUX

Physical port, AUX UART

MCU

Internal port, Reserved (BG772A-GL, BG952A-GL)

EMUX0

Virtual port 0, EMUX function

EMUX1

Virtual port 1, EMUX function

EMUX2

Virtual port 2, EMUX function

EMUX3

Virtual port 3, EMUX function

ACM0

Virtual port 0, USB interface

ACM1

Virtual port 1, USB interface

ACM2

Virtual port 2, USB interface

ACM3

Virtual port 3, USB interface

<function>

String type.

AT_PPP

AT client 0 / PPP data transfer

AT_1

AT client 1

AT_2

AT client 2

GNSS_NMEA

GNSS NMEA output

SFP_LOGGER

FW log messages, Debug mode

	FW_LOG	LTE log messages, Debug mode
	MAC_VIA_MAP	LTE log messages, Debug mode, USB/EMUX mode
	ALTCOM	Internal function, Reserved (BG772A-GL, BG952A-GL)
	None	No function is allocated to <port>
<err>	Integer type. Error code. See Chapter 3 for more information.	

2.1.2.21. AT+QCFG="uartmapping/profile" User Port Mapping Profile

The command reads/writes the parameter values in module port profile mapping configurations. The current user mapping configuration of the selected port profile can be queried with **AT+QCFG="uartmapping"**.

AT+QCFG="uartmapping/profile" User Port Mapping Profile	
Write Command AT+QCFG="uartmapping/profile"[,<profile>]	Response If the optional parameter is omitted, query the current configuration: +QCFG: "uartmapping/profile",<profile> OK If the optional parameter is specified, set the user port profile mapping configuration OK If there is an error related to ME functionality: ERROR If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<profile>	Integer type.	Delta changes only for modified settings
	<u>0</u> User profile #0	Mode USB+DEBUG
		DBG None
		MAIN AT1
		AUX None
		ACM0 AT/PPP
		ACM3 GNSS

1	User profile #1	Mode	USB+DEBUG
		DBG	GNSS_NMEA
		MAIN	AT/PPP
		AUX	AT1
		ACM0	None
		ACM3	None

2.1.2.22. AT+QCFG="urc/delay" Set URC Output Time

The command queries and sets URC output time.

AT+QCFG="urc/delay" Set URC Output Time

Write Command AT+QCFG="urc/delay"[,<enable>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "urc/delay",<enable> OK If the optional parameter is specified, set URC output time: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<enable>	Integer type. URC output time. <u>0</u> Output URC when ring indication pulse starts. 1 Output URC when ring indication pulse ends (effective only when <typeRI>="pulse" . See AT+QCFG="urc/ri/ring" , AT+QCFG="urc/ri/smsincoming" and AT+QCFG="urc/ri/other" for more information).
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.23. AT+QCFG="urc/ri/other" Set MAIN_RI Behavior for Other URCs

The command queries and sets the behavior of the MAIN_RI pin when there are other URCs.

AT+QCFG="urc/ri/other" Set MAIN_RI Behavior for Other URCs

Write Command

AT+QCFG="urc/ri/other"[,<typeRI>,<pulse_duration>,<pulse_count>]]]

Response

If the optional parameters are omitted, query the current setting:

+QCFG: "urc/ri/other",<typeRI>,<pulse_duration>,<pulse_count>

OK

If any of the optional parameters is specified, set the MAIN_RI behavior for other URCs.

OK

If there is an error related to ME functionality:

+CME ERROR: <err>

If there is any other error:

ERROR

Maximum Response Time

300 ms

Characteristics

The command takes effect immediately.
The configurations are saved automatically.

Parameter

<typeRI>	String type. MAIN_RI behavior for other URCs. "off" No change. MAIN_RI pin stays inactive (high). "pulse" Pulse. Pulse width determined by <pulse_duration> .
<pulse_duration>	Integer type. Pulse width. Range: 1–2000. Default: 120. Unit: ms. Parameter is valid only when <typeRI>="pulse" .
<pulse_count>	Integer type. Pulse count. Parameter is valid only when <typeRI>="pulse" . Range: 1–5. Default: 1. Interval between two pulses equals <pulse_duration> .
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.24. AT+QCFG="urc/ri/ring" Set MAIN_RI Behavior for RING URC

The command queries and sets the behavior of MAIN_RI pin when **RING** URC is reported to indicate an incoming call.

The sum of **<active_duration>** and **<inactive_duration>** determines the interval of RING indication for an incoming call.

AT+QCFG="urc/ri/ring" Set MAIN_RI Behavior for RING URC

Write Command	Response
AT+QCFG="urc/ri/ring"[,<typeRI>[,<pulse_duration>[,<active_duration>[,<inactive_duration>[,<ring_no_disturbing>[,<pulse_count>]]]]]]	If the optional parameters are omitted, query the current setting: +QCFG: "urc/ri/ring",<typeRI>,<pulse_duration>,<active_duration>,<inactive_duration>,<ring_no_disturbing>,<pulse_count> OK If any of the optional parameters is specified, set the behavior of MAIN_RI pin when RING URC is reported. OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<typeRI>	String type. MAIN_RI behavior when RING URCs are reported.
"off"	No change. MAIN_RI pin stays inactive (high).
" <u>pulse</u> "	Pulse. Pulse width is determined by <pulse_duration> .
"always"	Change to active. MAIN_RI behavior can be restored to inactive with AT+QIRI (See document [1] for more information).
"auto"	When a RING URC is reported to indicate an incoming call, MAIN_RI pin changes to active state and remains active. Answering or hanging up the incoming call changes MAIN_RI pin state to inactive.

	"wave"	When a RING URC is reported to indicate an incoming call, MAIN_RI pin outputs a square wave. Both <active_duration> and <inactive_duration> are used to set the square wave. Answering or hanging up the incoming call changes the MAIN_RI pin state to inactive.
<pulse_duration>		Integer type. Pulse width. Range: 1–2000. Default: 120. Unit: ms. Parameter is valid only when <typeRI>="pulse" .
<active_duration>		Integer type. Active duration of the square wave. Range: 1–10000. Default: 1000. Unit: ms. Parameter is valid only when <typeRI>="wave" .
<inactive_duration>		Integer type. Inactive duration of the square wave. Range: 1–10000. Default: 5000. Unit: ms. Parameter is valid only when <typeRI>="wave" .
<ring_no_disturbing>		String type. Set whether the MAIN_RI pin behavior could be disturbed. Parameter is valid only when <typeRI>="auto" or "wave" . For example, when <typeRI>="wave" , to prevent disturbance of the square wave by other URCs (including SMS-related URCs), set <ring_no_disturbing> set to "on".
	<u>"off"</u>	MAIN_RI pin behavior can be changed by other URCs when an incoming call is ringing.
	"on"	MAIN_RI behavior cannot be changed by other URCs when an incoming call is ringing.
<pulse_count>		Integer type. Pulse count. Parameter is valid only when <typeRI>="pulse" . Range: 1–5. Default: 1. Interval between two pulses is equal to <pulse_duration> .
<err>		Integer type. Error code. See Chapter 3 for more information.

2.1.2.25. AT+QCFG="urc/ri/smsincoming" Set MAIN_RI Behavior for Incoming SMS URCs

The command queries and sets the behavior of the MAIN_RI pin for incoming SMS URCs. Incoming SMS URCs include **+CMTI**, **+CMT**, **+CDS**, and **+CBM** (See **document [1]** for details).

AT+QCFG="urc/ri/smsincoming" Set MAIN_RI Behavior for Incoming SMS URCs

Write Command	Response
AT+QCFG="urc/ri/smsincoming"[,<typeRI>,<pulse_duration>,<pulse_count>]]]	If the optional parameters are omitted, query the current setting: +QCFG: "urc/ri/smsincoming",<typeRI>,<pulse_duration>,<pulse_count>
	OK
	If any of the optional parameters is specified, set the MAIN_RI behavior in case of incoming SMS URCs: OK

	If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<typeRI>	String type. MAIN_RI behavior for incoming SMS URCs. "off" No change. MAIN_RI pin stays inactive (high). "pulse" Pulse. Pulse width is determined by <pulse_duration>. "always" Change to active. MAIN_RI behavior can be restored to inactive with AT+QIR (See document [1] for more information).
<pulse_duration>	Integer type. Pulse width. Range: 1–2000. Default: 120. Unit: ms. Parameter is valid only when <typeRI>="pulse".
<pulse_count>	Integer type. Pulse count. Parameter is valid only when <typeRI>="pulse". Range: 1–5. Default: 1. Interval between two pulses equals <pulse_duration>.
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.26. AT+QCFG="usb" Activate/Deactivate USB Interface

The command activates/deactivates the USB Interface. To use the USB interface, the module should be connected via the USB cable to the host computer. Following the boot process, the module provides access to 3 additional ports.

AT+QCFG="usb" Activate/Deactivate USB Interface

Write Command AT+QCFG="usb",<enable>	Response If the optional parameter is omitted, query the current setting: +QCFG: "usb",<enable> OK If the optional parameter is specified, activate/deactivate USB interface: OK If there is an error related to ME functionality: +CME ERROR: <err>
---	---

	If there is any other error: ERROR
Maximum Response Time	1500 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved automatically.

Parameter

<enable>	Integer type. Whether to activate/deactivate USB Interface. 0 Deactivate 1 Activate 2 Activate USB and ECM
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.27. AT+QCFG="usb/urcport" Set URC Output Port to USB Interface Mode

The command queries and sets the URC output port to USB interface mode.

AT+QCFG="usb/urcport" Set URC Output Port to USB Interface Mode	
Write Command AT+QCFG="usb/urcport",<URC_port>	Response If the optional parameter is omitted, query the current setting: +QCFG: "usb/urcport",<URC_port> OK If the optional parameter is specified, set the URC output port to USB interface mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<URC_port>	Integer type. URC output port in USB Interface mode. <u>0</u> All ports 1 Virtual port 1 2 Virtual port 2 3 Virtual port 3
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.28. AT+QCFG="cmux/urcport" Set URC Output Port to CMUX Mode

The command queries and sets the URC output port to CMUX mode.

AT+QCFG="cmux/urcport" Set URC Output Port to CMUX Mode

Write Command AT+QCFG="cmux/urcport"[,<URC_port>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "cmux/urcport",<URC_port> OK If the optional parameter is specified, set the URC output port to CMUX mode: OK If there is an error related to ME functionality: +CME ERROR: <err> If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<URC_port>	Integer type. URC output port in CMUX mode. 0 All ports 1 Virtual port 1 <u>2</u> Virtual port 2 3 Virtual port 3
<err>	Integer type. Error code. See Chapter 3 for more information.

2.1.2.29. AT+QCFG="fast/poweroff" Control Fast Shutdown Function

The Command enables/disables fast shutdown function triggered by the specified pin.

AT+QCFG="fast/poweroff" Control Fast Shutdown Function

Write Command AT+QCFG="fast/poweroff" [<pin>,<enable>]	Response If the optional parameters are omitted, query the current setting: +QCFG: "fast/poweroff",<pin>,<enable> OK If the optional parameters are specified, enable/disable fast shutdown function triggered by the specified pin: OK If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<pin>	Integer value. Number of the GPIO pin configured in input mode with a pull-up resistor, which can trigger a fast shutdown upon detecting a falling edge. Corresponding pin number for BG95xA-GL series: 85 Corresponding pin number for BG77xA-GL series: 57 Corresponding pin number for BG950S-GL module: 66
<enable>	Integer value. Enable/disable fast shutdown. 0 Disable 1 Enable

2.1.2.30. AT+QCFG="dhclkrefresh" Turn Off PMU LP Refresh Based on Current Temperature

This command enables the behavior of turning off PMU LP (Power Management Unit Low Power) refresh till the device exits low power mode if the current temperature is not more than the defined temperature threshold.

AT+QCFG="dhclkrefresh" Turn Off PMU LP Refresh Based on Current Temperature

Write Command AT+QCFG="dhclkrefresh",<enable>,<upper_temp_limit>	Response OK If there is any other error: ERROR
Maximum Response Time	300 ms
Characteristics	The configuration is saved automatically.

Parameter

<enable>	Integer value. Enable/disable the behavior of turning off PMU LP refresh. 0 Disable 1 Enable
<upper_temp_limit>	Integer value. Upper limit of the temperature threshold. Range: -40 to +85. Unit: °C.

Example

```

AT+QCFG="dhclkrefresh",1,5 //Enable the behavior when the module is below 5 °C
OK
AT+QCFG="dhclkrefresh",1,85 //Enable the behavior over the entire temperature range
OK
AT+QCFG="dhclkrefresh",0,85 //Disable the behavior using the last applicable value as
                             <upper_temp_limit>
OK

```

3 Summary of +CME ERROR Codes

Final result code **+CME ERROR: <err>** indicates an error related to mobile equipment or network. The operation is similar to the **ERROR** result code.

<err> values are mostly used by common message commands. The following table lists most general and GPRS-related **ERROR** codes. The corresponding **ERROR** codes for some GSM protocol failure causes described in GSM specifications are not included.

Table 2: Summary of +CME ERROR Codes

Numeric <err> Value	Verbose <err> Value
0	phone failure
1	no connection to phone
2	phone-adaptor link reserved
3	operation not allowed
4	operation not supported
5	PH-SIM PIN required
6	PH-FSIM PIN required
7	PH-FSIM PUK required
10	SIM not inserted
11	SIM PIN required
12	SIM PUK required
13	SIM failure
14	SIM busy
15	SIM wrong

16	incorrect password
17	SIM PIN2 required
18	SIM PUK2 required
20	memory full
21	invalid index
22	not found
23	memory failure
24	text string too long
25	invalid characters in text string
26	dial string too long
27	invalid characters in dial string
30	no network service
31	network timeout
32	network not allowed - emergency calls only
40	network personalization PIN required
41	network personalization PUK required
42	network subset personalization PIN required
43	network subset personalization PUK required
44	service provider personalization PIN required
45	service provider personalization PUK required
46	corporate personalization PIN required
47	corporate personalization PUK required

4 Appendix References

Table 3: Related Document

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL&BG950S-GL_AT_Commands_Manual
[2] Quectel_BG77xA-GL&BG95xA-GL&BG950S-GL_PSM_Application_Note

Table 4: Terms and Abbreviations

Abbreviation	Description
AP	Application Processor
APN	Access Point Name
CDS	Common Data Service
CE	Coverage Enhancement
ECM	Embedded Connection Manager
eDRX	Extended Discontinuous Reception
EMM	EPS Mobility Management
EMUX	Multiplexing
eMTC	Enhanced Machine-Type Communication
ePCO	Extended Protocol Configuration Options
GPIO	General-Purpose Input/Output
GSM	Global System for Mobile Communications
IMSI	International Mobile Subscriber Identity
JSON	JavaScript Object Notation

LED	Light Emitting Diode
LTE	Long-Term Evolution
LwM2M	Lightweight M2M
ME	Mobile Equipment
NB-IoT	Narrowband Internet of Things
PCO	Protocol Configuration Options
PIN	Personal Identification Number
PS	Packet Switched
PSM	Power Saving Mode
PUK	PIN Unlock Key
RAT	Radio Access Technology
RRC	Radio Resource Control
SIB	System Information Block
SMS	Short Message Service
TA	Terminal Adapter
TAU	Tracking Area Update
URC	Unsolicited Result Code
(U)SIM	(Universal) Subscriber Identity Module